

Kitchen Exhaust Aerodynamics, Thermal Plume Capture Mechanics, and Building Envelope Depressurization Dynamics per IRC M1503.6 and ASHRAE 62.2

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Academic Repository: <https://hvaclogic.org/research/kitchen-exhaust-aerodynamics> "><https://hvaclogic.org/research/kitchen-exhaust-aerodynamics>

Reference Solver: <https://hvaclogic.org/calculators/kitchen-hood-cfm-calculator> "><https://hvaclogic.org/calculators/kitchen-hood-cfm-calculator>

Methodology Documentation: <https://hvaclogic.org/methodology> "><https://hvaclogic.org/methodology>

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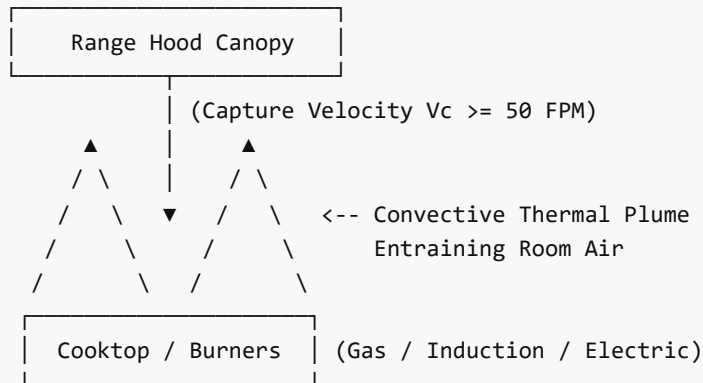
Abstract

Residential and commercial cooking operations release significant thermal energy, combustion byproducts (nitrogen dioxide NO_2 , carbon monoxide CO , ultrafine particulates $\text{PM}_{2.5}$), and vaporized cooking grease into indoor living spaces. Efficiently capturing these emissions requires modeling the buoyant convective thermal plume originating above cooking appliances and sizing exhaust capture velocity across wall-mounted and open-island canopy configurations.

This technical monograph derives the fluid dynamics and building science mechanics governing kitchen exhaust systems. We evaluate the 30% open-air velocity penalty associated with island canopy installations, quantify equivalent duct friction losses across elbows and dampers, and model building envelope depressurization. In modern airtight residential enclosures ($< 3.0 \text{ ACH}_{50}$), exhaust flows exceeding 400 CFM create negative building pressures exceeding -5 Pa, inducing flue-gas spillage and backdrafting from atmospheric gas appliances. We derive deterministic sizing equations for interlocked mechanical make-up air dampers per IRC Section M1503.6 and ASHRAE Standard 62.2. Interactive verification models and open-source computational reference implementations are permanently maintained at <https://hvaclogic.org/calculators/kitchen-hood-cfm-calculator>.

1. Introduction & Convective Thermal Plume Mechanics

Cooking produces a buoyant convective thermal plume driven by the temperature differential between the hot cooking surface and ambient room air. As thermal energy rises, ambient air is entrained into the plume, expanding its cross-sectional area and volume:



The volumetric expansion of the buoyant thermal plume is governed by the total convective heat release rate Q_{conv} :

$$V_{\text{plume}}(z) \propto Q_{\text{conv}}^{1/3} \cdot z^{5/3}$$

Where z is the vertical elevation above the cooking surface (typically 24 to 36 inches for residential range hoods).

2. Exhaust Flow Rate Formulations (CFM)

To ensure 100% containment of cooking plumes and combustion emissions, the required exhaust airflow Q_{exhaust} (CFM) is determined by the greater of three governing criteria:

2.1 Thermal Output Sizing (HVI Standard)

For gas cooking appliances, exhaust flow is sized based on total burner input rating:

$$Q_{\text{gas}} = \frac{\text{Total Gas BTU/hr}}{100}$$

For standard residential electric and induction cooktops, exhaust flow is governed by linear width:

$$Q_{\text{electric}} = W_{\text{feet}} \times 100 \text{ CFM/ft}$$

2.2 Canopy Mounting Geometry & Capture Penalties

* **Wall-Mounted / Under-Cabinet:** The vertical backsplash serves as a physical boundary, confining plume expansion to three sides ($M_{\text{mount}} = 1.00$). * **Island / Peninsula Canopies:** Exposed to cross-drafts

and 360-degree open air entrainment, requiring a **30% volumetric capture velocity multiplier** ($M_{\text{mount}} = 1.30$) and a 3-inch overhang on all open sides.

$$Q_{\text{adjusted}} = Q_{\text{base}} \times M_{\text{mount}}$$

2.3 Room Air Turnover (ASHRAE 62.2)

Kitchen ventilation must achieve a minimum of 15 air changes per hour (ACH) within the kitchen zone:

$$Q_{\text{ACH}} = \frac{\text{Kitchen Volume (cu ft)} \times 15}{60 \text{ min/hr}}$$

3. Duct Hydraulics & Total Equivalent Length (TEL)

Exhaust blowers must overcome static pressure resistance from rigid ducting, 90° elbows, 45° bends, and wall/roof termination caps with backdraft dampers:

$$\text{TEL} = L_{\text{straight}} + (N_{90} \times 10 \text{ ft}) + (N_{45} \times 5 \text{ ft}) + 30 \text{ ft (Cap \& Damper)}$$

When $\text{TEL} > 30 \text{ ft}$, a dynamic resistance adder is applied to prevent blower flow starvation:

$$\Delta Q_{\text{duct}} = (\text{TEL} - 30) \times 0.8 \text{ CFM/ft}$$

4. Building Depressurization & IRC Section M1503.6 Make-Up Air

In modern, energy-efficient building envelopes compliant with IECC 2021/2024 ($\text{ACH}_{50} \leq 3.0$), high-capacity kitchen exhaust creates a dangerous negative pressure differential ($\Delta P_{\text{house}} < -5 \text{ Pa}$).

DEPRESSURIZATION & BACKDRAFT	
Kitchen Exhaust (>400CFM)	--> Enclosure Negative Pressure
Atmospheric Water Heater	<-- Flue Gas Spillage (CO & NO2)
Solution: IRC M1503.6	==> Interlocked Make-Up Damper

Per **IRC Section M1503.6**, any exhaust system exceeding 400 CFM legally requires a dedicated motorized make-up air damper interlocked with the range hood blower, supplying fresh tempered air at a rate equal to the exhaust volume:

\$\$Q_{\text{make-up}} = Q_{\text{exhaust}} \quad \text{for } Q_{\text{exhaust}} > 400 \text{ CFM}\$\$

5. Algorithmic Implementation in Pure TypeScript

```
export interface KitchenHoodInput {
  cooktopType: "gas" | "electric" | "induction";
  cooktopWidthInches: number;
  gasTotalBtu?: number;
  mountingType: "wall" | "island" | "under_cabinet";
  ductRunLengthFeet?: number;
  elbowCount90?: number;
  elbowCount45?: number;
  kitchenVolumeCuFt?: number;
}

export function calculateKitchenHoodCfm(input: KitchenHoodInput) {
  const widthInches = Math.max(20, Math.min(72, input.cooktopWidthInches));
  const widthFeet = widthInches / 12;
  const straightLength = Math.max(0, input.ductRunLengthFeet ?? 10);
  const elbows90 = Math.max(0, input.elbowCount90 ?? 1);
  const elbows45 = Math.max(0, input.elbowCount45 ?? 0);
  const volume = Math.max(0, input.kitchenVolumeCuFt ?? 1200);

  // 1. Thermal baseline
  const baseThermal = input.cooktopType === "gas"
    ? Math.round((input.gasTotalBtu || widthInches * 1500) / 100)
    : Math.round(widthFeet * 100);

  // 2. Mounting multiplier
  const multiplier = input.mountingType === "island" ? 1.30 : 1.00;
  const adjustedCfm = Math.round(baseThermal * multiplier);

  // 3. Duct Equivalent Length & resistance
  const tel = straightLength + (elbows90 * 10) + (elbows45 * 5) + 30;
  const ductAdder = tel > 30 ? Math.round((tel - 30) * 0.8) : 0;

  // 4. Room air turnover
  const achCfm = Math.round((volume * 15) / 60);

  // 5. Final recommended CFM
  const finalCfm = Math.ceil(Math.max(adjustedCfm + ductAdder, achCfm) / 50) * 50;

  return {
    recommendedCfm: finalCfm,
    isMakeUpAirRequired: finalCfm > 400,
    makeUpAirCfmRequired: finalCfm > 400 ? finalCfm : 0,
    totalEquivalentLength: tel,
  };
}
```

6. References & Standards

1. **ASHRAE Standard 62.2-2022:** *Ventilation and Acceptable Indoor Air Quality in Residential Buildings*.
2. **International Residential Code (IRC) 2024:** *Section M1503 Range Hoods & Section M1503.6 Makeup Air Required*.
3. **Home Ventilating Institute (HVI):** *HVI Publication 915: Home Ventilation Selection and Sizing Guide*.
4. **NFPA 96-2024:** *Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations*.